A Symmetrical Robotic Arm Design Approach with Stereo-vision Ability for CubeSats

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Abstract— this paper introduces a design of a robotic system that can be mounted on a CubeSat. Robotic arms should be small enough to be mounted on a size as 30x10x10 cm or smaller. Meanwhile, robotic arms should provide sufficient space for their actuators, cameras and sensors. The robotic design has one camera on each arm to be used for stereo-vision ability. Velocity and position feedback are desired to be used for controlling the robotic arm's position and minimizing the speed of the arms' motion to reduce the disturbance on attitude control subsystem during arm motions. Each arm provides 5 degrees of freedom. For control design of the system, the Denavit-Hartenberg convention is used for forward and backward kinematics modeling, therefore the position and the orientation of the camera can be calculated after receiving a command from the ground station or ready mission parameters from the memory. By using forward kinematics matrices, the Jacobian matrix will be presented for the system, and it will be used to calculate the inertial tensor matrix which includes the effects of the mass and inertia. After calculating the inertial tensor matrix, Euler-Lagrangian method will be used for dynamic modeling which includes the nonlinear coriolis and centrifugal effects with the inertial forces. The dynamic equation of motion for two arms which includes inertial forces, coriolis and centrifugal effects, friction forces and gravity will be presented for space and the micro gravity environment. Moreover, spin-stabilized satellite motion effect is added to robotic arms' the dynamic equation of the motion. The dynamic equation is used with the proportional-integral coefficients for modeling the control of the prismatic links' actuators chosen for the arms.

Keywords—robotic arm; space robotics, inspection mission;

I. INTRODUCTION

Space robotics is the development of robots which are capable of surviving in the space environment conditions, at least for a time period, to perform exploration, assembly, inspection, maintenance, servicing, construction or other tasks. To perform a task while being controlled by a human operator, a space robot will be experiencing either non-negligible speed-of-light delays as in the Mars exploration or essentially zero speed-of light delay as in the Space Shuttle robot arm [1]. From space robotics perspective, it can be said that a robot is a goal oriented machine that can sense its environment, get information about it and use that information to plan its actions and act. These actions might be moving its arm, moving to another position, interacting with some objects or someone or waiting for a new task. In order to plan and achieve these actions, a robot should have a certain amount of intelligence. For the case of Mars exploration, non-negligible speed-of-light delays point out the importance of the autonomous capabilities of a space robot since it cannot change the environment conditions. If the robot's environment is not changing, it can make decisions more accurately; however, the information about the environment from the sensors' data should be understood and used efficiently in order to perform the tasks and survive while accomplishing them.

Since robots can perform many risky missions, which humans are not allowed to do, without being tired and less expensive than a manned mission, space robots became attractive for many researches. The important issues for a space robot are:

- Surviving and operating in a harsh environment such as intense heat or cold, radiation, vacuum etc.
- Using arms and tools to perform tasks and interact with its environment and worksite elements accurately and safely.
- Allowing a time delayed control
- Moving quickly and safely to reach a desired position without collisions

II. INSPECTION MISSIONS

Robotic systems proved their efficiency in space as well as on Earth while they played significant role on many fields. As one of the research areas of space robotics, orbital robotics missions have possible objectives as maintenance, human-robot cooperation, assembly and inspection. Robotic based on-orbit servicing spacecraft are internationally gaining attention for emergency situations and also economic reasons [2]. Disassembling a working component of a satellite which has a malfunction occurred in another subsystem component and using that component on another satellite can reduce the cost of a mission. In the meanwhile, understanding the reason of the malfunction and repairing the satellite is more efficient than launching a new one. By inspection missions, the coverage of exterior structures is important as well as mobility. Reaching a useful perspective to monitor the required surfaces is essential.

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The important issues for an inspection mission in outer space:

- Adapting to changing orbital lighting conditions
- Being able to reach a sufficient viewpoint to have a better perspective
- Minimizing the attitude disturbances

III. STRUCTURAL AND MECHANICAL DESIGN

In inspection missions, a robotic arm should provide sufficient mobility, and be able to reach a pose with an efficient perspective in order to monitor either satellite's or another body's surface. To design a robotic system that can be mounted on a CubeSat, robotic arms should be small enough to be mounted on a size as 30x10x10 cm or smaller which varies from 1U to 3U. Meanwhile, robotic arms should provide sufficient space for their actuators, cameras and sensors. Each robotic arm has one camera on each arm to be used for stereo-vision ability. The actuators on arms are used for positioning of the cameras, and the actuators on the tips are used for orientation of the cameras.

In design process, the robotic arms were desired to occupy most of the space on the top of CubeSat where they are mounted, in order to minimize the effect of vibration. It is similar to if they were slotted as shown in Figure 01. The mechanical design objectives can be written as:

- To have a symmetrical dual-arm robotic system
- To have sufficient degrees of freedom
- To maximize the working space
- To minimize inertia disturbance on the attitude determination and control subsystem
- To have enough space for actuators, sensors and cameras
- To occupy minimum space from CubeSat
- To minimize the risk of contact or collision of the arms

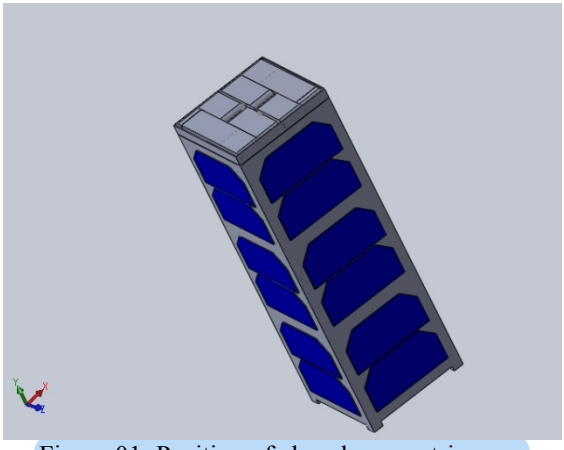


Figure 01: Position of closed symmetric arms

Each part of the robotic arm is designed by examining commercially available hardware choices. Even though the chosen components of the hardware which were taken as an example were not suitable for outer space conditions, their components can be replaced with new components which have materials those can endure space conditions. Moreover, they can be produced and ordered specially for specific mission types. Typically, this includes conditions of microgravity, hard vacuum and extreme temperature variations.

Inertia properties of spacecraft can change in orbit for many reasons, therefore attitude determination and control subsystems have to adept these changes. Having 5 degrees of freedom provides an ability to reach a point by different configurations of the robotic arms; however, it is important to consider that opening the arms wider will change the initial kinetic energy of the system.

A. The Advantages of Symmetrical Dual-Arm Robotic System

- Two cameras can take images of a larger exterior area and the earth background in a single shot.
- Having two arms can reach more space at the same time shortens the mission time.
- When one arm breaks down, the other one can carry on the inspection.
- It minimizes the disturbance on the inertia of the main body.
- The system can have a secondary purpose that is to assist the determination of the attitude by digital images of the Earth since the position and orientation of the cameras precisely known with respect to the CubeSat.
- It can be used as a secondary de-orbiting system by opening the arms to decrease the kinetic energy in the end of life.

IV. KINEMATICS

Kinematics is the branch of classical mechanics that studies the motion of points, bodies and systems of bodies without consideration of its mass or the forces acting on them [3]. A robotic arm comprises a chain of mechanical links and joints which allow actuators to be mounted on its structure. The kinematics of five degrees of freedom arm is expressed by using Denavit Hartenberg convention. First, the Denavit Hartenberg parameter tables are presented:

TABLE 1: The given values and variables for the right arm.

	a	A	θ	b	Ready-state value
R ← O	0	0°	0°	0	-
0 ← 1	0	90°	θ_1	L_2	-90°
1 ← 2	L_o	0°	θ_2	0	90°
2 ← 3	0	90°	θ_3	$-L_3$	90°
3 ← 4	0	90°	180	b_4	$L_4 + L_5$↗
4 ← 5	L_1	-90°	θ_5	L_6	90°

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TABLE 2: The given values and variables for the left arm.

	a	α	θ	b	Ready-state value
R ← O	0	0°	0°	0	-
0 ← 1'	0	90°	$\theta_{1'}$	L_2	-90°
1' ← 2'	L_o	0°	$\theta_{2'}$	0	90°
2' ← 3'	0	90°	$\theta_{3'}$	L_3	90°
3' ← 4'	0	90°	0°	$b_{4'}$	$L_4 + L_5$↗
4' ← 5'	L_1	-90°	$\theta_{5'}$	L_6	90°

The values of the distances between axes and the lengths of the robotic arm parts:
L_o = 43.50 mm
L_1 = 33.50 mm
L_2 = 5 mm
L_3 = 34.60 mm
L_4 = 26.55 mm
L_5 = from 0 mm to 56.95 mm
L_6 = 23.05 mm
a_1 = 11.55 mm

For inverse kinematic solution, where the position and orientation of the cameras are defined with B_x, B_y, B_z and θ_5, required joint variables will be calculated as:

$$\theta' = atan2(a_1, \sqrt{{B_x}^2 + {B_y}^2 - {a_1}^2}) \; - atan2(B_y, B_x) \quad (1)$$

$$\theta_1 = -\frac{\pi}{2} - \theta' \quad (2)$$

For the dual arm robotic system, it is desired that two arms are moving symmetrically. Therefore, when the desired camera coordinates and angle are sent by the ground station or received from the memory, inverse kinematics provides a solution to determine the required values for being used to control the actuators.

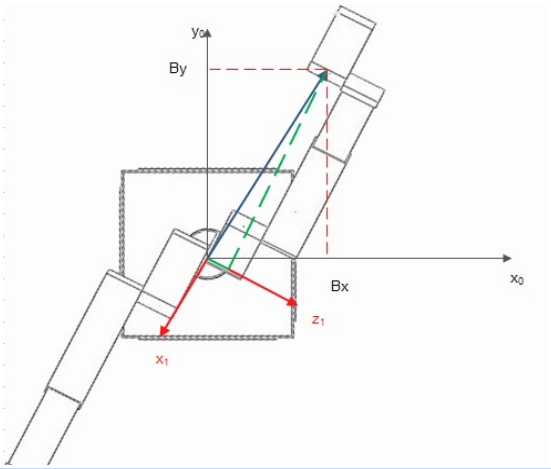


Figure 02: The view of the symmetric arms from above

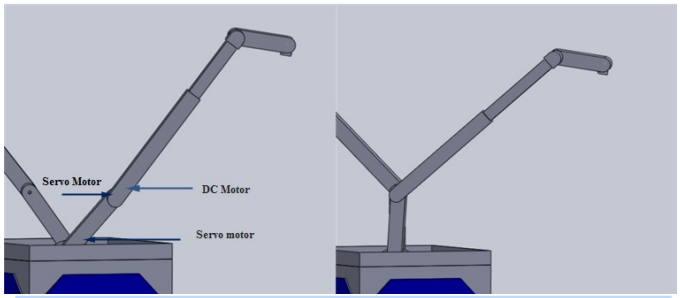


Figure 03: The comparement between two positions to reach the same point

By using the algorithm programmed on Matlab, the program calculates the possible points for the camera origin. After running the first part of the program, if the given point is not in the workspace area, it says "Not possible to reach". Otherwise, it starts the calculation for θ_3 and b_4 as well as θ_1.

$$\theta' = atan2\ (B_z - 48.50, \sqrt{{B_x}^2 + {B_y}^2 - {a_1}^2}) \quad (3)$$

$$\theta_3 = \frac{\pi}{2} + \theta' \quad (4)$$

$$\theta_{3'} = \frac{\pi}{2} - \theta \quad (5)$$

$$b_4 = b_{4'} = \frac{\sqrt{{B_x}^2 + {B_y}^2 - {a_1}^2}}{\sin(\theta')} - 88.5 \quad (6)$$

If it is possible to reach to the desired position, θ_2 angle is directly chosen as 90° in order to hold the arms' first part close to the vertical axis z_R. Since those parts of the robotic arms, which are neighboring the shoulder, contain two servo motors and a DC motor in a close position for each arm, the inertia distribution is considered.

As shown in Figure 03, there are different sets of joint variables to reach the same pose for the camera as mentioned earlier. Due to mechanical limits on the joint variables, most of the joint poses are not available. Meanwhile, inertia considerations also help to limit these configuration choices, and it becomes manageable.

V. Dynamics and Control

A. Dynamics

The Jacobian matrix, which is written as J, is one of the most important quantities in the analysis and control of robot motion. Since the Jacobian matrix encodes relationships between velocities and positions, it will be used in the derivation of the dynamic equations of motion and the transformation of forces and torques from the end-effector to the manipulator joints. It is used for derivation of the inertial tensor matrix (7).

$$i = 1,2,3,4,5 \quad E_i = m_i[{J_L^{(i)}}^T * J_L^{(i)}] + [{J_A^{(i)}}^T * I_i * J_A^{(i)}] \quad (7)$$

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The inertia tensor matrix shows all mass and inertial properties of the robotic systems, and it is very important to be derived for understanding the dynamics of a robotic system since the inertia of the manipulator should be surpassed in order to move the arms.

The general form of the kinetic energy for a serial link manipulator is given as:

$$T = \frac{1}{2}\{\dot{q}^T * E * \dot{q}\} \quad (8)$$

The Lagrangian function expression can be written as:

$$L = T - U \quad (9)$$

For a robotic arm system with five degrees of freedom, the Lagrange equation includes five generalized coordinates and five generalized velocities.

The generalized form of the torque which an actuator should provide in order to move its joint can be given as [4]:

$$\Gamma_i = \frac{d}{dt}\left\{\frac{\partial L}{\partial \dot{q}_i}\right\} - \frac{\partial L}{\partial q_i} \qquad where\ \ i = 1,2,3,4,5 \quad (10)$$

The generalized dynamic equations of the robotic arms contain inertial forces, centripetal and coriolis torques, friction forces and gravity loading where q, $\dot{q}$ and $\ddot{q}$ are respectively the generalized joint coordinates, velocities and accelerations.

If the CubeSat is a spin-stabilized satellite, the rotational motion of the satellite should be added to dynamic equation of the motion in order to obtain an efficient model since all parts of the robotic arms would be receiving the effect of the moment of a revolving body. After modification of the rotational motion equations of an airplane as [5]:

$$\sum \Delta M = \frac{dH}{dt}\Big|_E \quad (11)$$

The momentum ΔM can be derived from (11) with the angular momentum H. Moreover, the angular velocity w_z will be same for all parts of the robotic arms, however, the distance between the gravity centers and reference axis' origin $\vec{r}$ will be changing over time.

$$d\vec{H} = \vec{r}x\left(\vec{w}x\vec{r}\right).dm \quad (12)$$

After using (11) and (12), the equations of the angular motion about the axes of x, y and z are derived. If there is only the constant angular velocity w_z, it can be simplified and written as;

$$\sum \Delta M = -w_z^2\, I_{xz} \quad (13)$$

The dynamic equation of the motion provides an understanding of the terms such as how motion of one joint exerts disturbance on the other joints, how inertia and gravity load varies with the links' poses. After deriving the equation of the motion it can be used to make a control design approach in order for the cameras of the robotic arms follow a desired Cartesian trajectory and reach their desired position.

B. Control

Servo motors are chosen for most of the actuators which are needed; however, the second part of the arm contains prismatic link which makes the movement of either contraction or extension by controlling a DC motor with a worm gear.

For the control of the servo motors a control circuit, an incoming signal and a potentiometer is needed. The potentiometer, which controls the angle of the servo motor, allows the control circuitry to monitor the current angle of the servo motor.

To control the DC motors on the prismatic links of the arms, sensors are needed to estimate their initial positions. The feed-back signal for the servo motors can be taken from its position indicator encoders' wire; however, the DC motors need a sensor similar to linear absolute encoder or incremental encoder. Since incremental encoder uses light to sense during the entire motion, it is not desired to be used on the system to minimize the energy consumption. Meanwhile, the linear absolute encoder has the ability to sense its position to the reference point in case of system power-down and restart if the encoder changes its position during a power down situation. Also, it can sense position and speed accurately.

PI control is chosen for the DC motors; whereas the integral term considers the history of the error while the proportional term considers the current size of error in the controller calculation. It is desired for the robotic arms to move at a constant speed which is very low. Therefore, velocity and position feed-backs are needed when controlling the arm motions from one position to another. Integration can be considered as a continual summing that it sums the complete controller error over time, and as the arms' motions will disturb the attitude of the satellite, the satellite's motions will disturb the arms' control as well. Integral control will balance these disturbances. On the other hand, since the speed of the actuators will be constant and minimum, the derivative term wasn't added to the control term.

$$\gamma_4 = K_m A_1 U \quad (14)$$

$$U = K_p(\dot{\theta}_d - \dot{\theta}_o) + K_i\left(\int \dot{\theta}_d dt - \int \dot{\theta}_o dt\right) \quad (15)$$

It is shown that K_m is the motor constant which can be taken from the motor datasheet, A_1 is the operational amplifier, and U is the control signal which is derived from the error value between the desired value $\dot{\theta}_d$ and the measured value $\dot{\theta}_o$.

VI. Conclusions

This paper has an objective to present a design approach for a symmetrical dual-arm robotic system which can be mounted on a Cube-Sat. It was one of the optimum options for a robotic system which can operate on orbit, and it represents the dynamic model of five degrees of freedom robotic arms on a spin-stabilized satellite. For the next step of the project, the basic robotic arm movements can be listed, and a guideline can be established to select optimal perspectives to monitor needed surfaces or the Earth by moving slowly and safely to reach that desired position without collisions or inertial problems. The dynamics and control performances can be analyzed with Matlab/Simulink which is available. This robotic system also

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may have objectives as the secondary de-orbiting system by opening the arms as wide as possible to help the main de-orbiting system by decreasing the kinetic energy or determining the attitude by monitoring the Earth since the certain position and orientation of the camera is known with respect to the satellite for each arm.

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